Pharmacovigilance Internship Report

ADR Identification Basics — Educational Case Review

Dataset: case_scenarios.csv (18 synthetic case records)

Objective: Identify reported suspected drug–reaction pairs, review timing and follow-up information, and flag data-quality issues for further assessment.

Method

Each record was reviewed for the suspect product, reported reaction, onset after treatment began, reported seriousness, expectedness reference, dechallenge, rechallenge, and concomitant medicines. The CSV does not provide enough clinical detail to establish causality. 'Suspected ADR' below means a signal for assessment, not proof that the drug caused the event.

Summary by suspected product and reported reaction

Suspect product	Reported reaction	Cases (n)	Case IDs	Reported seriousness categories	Expectedness field
Amlodipine	Peripheral oedema	2	SYN-PV-004, SYN-PV-012	Medically important	Known
Amoxicillin	Rash	3	SYN-PV-001, SYN-PV-009, SYN-PV-017	Medically important	Known
Atorvastatin	Myalgia	3	SYN-PV-002, SYN-PV-010, SYN-PV-018	Hospitalisation, Not serious	Known
Ciprofloxacin	Tendon pain	2	SYN-PV-005, SYN-PV-013	Medically important	Known
Losartan	Fatigue	2	SYN-PV-008, SYN-PV-016	Not serious	Unexpected
Metformin	Nausea	2	SYN-PV-003, SYN-PV-011	Hospitalisation, Medically important	Known
Omeprazole	Headache	2	SYN-PV-007, SYN-PV-015	Hospitalisation, Medically important	Known
Sertraline	Dizziness	2	SYN-PV-006, SYN-PV-014	Hospitalisation, Medically important	Known

Case-by-case ADR identification table

Case ID	Age / sex	Suspect product	Reported reaction	Onset after start	Dechallenge	Rechallenge	Reported seriousness	Other medicines
SYN-PV-001	30 / Male	Amoxicillin (20 mg once daily)	Rash	17 day(s)	Negative	Positive	Medically important	None reported
SYN-PV-002	36 / Unknown	Atorvastatin (5 mg once daily)	Myalgia	1 day(s)	Unknown	Negative	Not serious	None reported
SYN-PV-003	34 / Female	Metformin (5 mg once daily)	Nausea	13 day(s)	Negative	Negative	Hospitalisation	Vitamin D
SYN-PV-004	61 / Male	Amlodipine (500 mg twice daily)	Peripheral oedema	21 day(s)	Positive	Negative	Medically important	None reported
SYN-PV-005	33 / Unknown	Ciprofloxacin (500 mg twice daily)	Tendon pain	15 day(s)	Unknown	Not done	Medically important	None reported
SYN-PV-006	31 / Unknown	Sertraline (20 mg once daily)	Dizziness	3 day(s)	Positive	Not done	Medically important	Paracetamol

Case ID	Age / sex	Suspect product	Reported reaction	Onset after start	Dechallenge	Rechallenge	Reported seriousness	Other medicines
SYN-PV-007	63 / Male	Omeprazole (500 mg twice daily)	Headache	16 day(s)	Negative	Negative	Medically important	Paracetamol
SYN-PV-008	36 / Male	Losartan (5 mg once daily)	Fatigue	2 day(s)	Negative	Not done	Not serious	Paracetamol
SYN-PV-009	78 / Unknown	Amoxicillin (5 mg once daily)	Rash	7 day(s)	Positive	Not done	Medically important	Vitamin D
SYN-PV-010	57 / Female	Atorvastatin (5 mg once daily)	Myalgia	21 day(s)	Unknown	Negative	Hospitalisation	Paracetamol
SYN-PV-011	23 / Unknown	Metformin (5 mg once daily)	Nausea	19 day(s)	Unknown	Positive	Medically important	None reported
SYN-PV-012	24 / Female	Amlodipine (5 mg once daily)	Peripheral oedema	4 day(s)	Unknown	Positive	Medically important	None reported
SYN-PV-013	75 / Female	Ciprofloxacin (500 mg twice daily)	Tendon pain	5 day(s)	Negative	Positive	Medically important	Vitamin D
SYN-PV-014	75 / Female	Sertraline (5 mg once daily)	Dizziness	9 day(s)	Negative	Negative	Hospitalisation	Paracetamol
SYN-PV-015	33 / Male	Omeprazole (20 mg once daily)	Headache	16 day(s)	Unknown	Positive	Hospitalisation	None reported
SYN-PV-016	66 / Unknown	Losartan (20 mg once daily)	Fatigue	10 day(s)	Positive	Negative	Not serious	None reported
SYN-PV-017	57 / Unknown	Amoxicillin (5 mg once daily)	Rash	21 day(s)	Unknown	Positive	Medically important	Vitamin D
SYN-PV-018	56 / Female	Atorvastatin (500 mg twice daily)	Myalgia	17 day(s)	Unknown	Positive	Hospitalisation	None reported

Clinical review and data-quality notes

Preliminary interpretation

The dataset contains 18 suspected drug–reaction reports covering eight product–reaction pairs: amoxicillin–rash (3 cases), atorvastatin–myalgia (3), metformin–nausea (2), amlodipine–peripheral oedema (2), ciprofloxacin–tendon pain (2), sertraline–dizziness (2), omeprazole–headache (2), and losartan–fatigue (2). These are reported events and should not be interpreted as incidence rates or confirmed causal relationships.

Assessment points

• **Timing:** Reported onset ranges from 1 to 21 days after treatment began. Timing may support consideration of a drug relationship but is not sufficient by itself.

• **Dechallenge:** Some cases report recovery or improvement after stopping the drug, while others report no recovery or unknown status. Interpret cautiously; follow-up duration and other clinical details are missing.

• **Rechallenge:** Several records say 'Positive'. This is important information to verify in the original case narrative. Do not recommend deliberately restarting a medicine to test causality.

• **Seriousness:** The file labels some cases as 'Hospitalisation' or 'Medically important'. These are reported classifications and have not been independently verified from source documents.

• **Expectedness:** The file marks losartan–fatigue cases as 'Unexpected' and most other reactions as 'Known'. The applicable product label/reference and version should be checked before a formal expectedness conclusion.

• **Concomitant medicines:** Paracetamol and vitamin D are listed in some records; the file does not provide enough detail to establish whether they contributed.

Data-quality issues requiring clarification

• Several dose/product combinations appear unusual or potentially erroneous in the supplied file (for example, amoxicillin 20 mg once daily; amlodipine 500 mg twice daily; omeprazole 500 mg twice daily; sertraline 20 mg once daily; and atorvastatin 500 mg twice daily). These must be checked against the source records; do not silently correct them.

• Some indication/product combinations appear inconsistent (for example, amoxicillin listed for anxiety, and sertraline listed for hypertension). Confirm the indication field and the intended suspect product.

• The narratives say 'documented indication' but do not specify medical history or detailed clinical findings. Missing facts should remain 'unknown', not be inferred.

• No laboratory results, relevant history, event outcome dates, alternative diagnoses, or full treatment dates are supplied. A formal causality score cannot be reliably assigned from this CSV alone.

Conclusion

The file identifies 18 cases for preliminary review. The reported drug–reaction pairs can be documented as suspected ADRs for assessment, but causality is not confirmed. Before final pharmacovigilance conclusions, verify the suspect drug, dose, indication, chronology, clinical details, seriousness, expectedness reference, and follow-up outcomes with the original source data.

Educational use: This report is based on a synthetic training dataset and is not a clinical diagnosis or a substitute for formal pharmacovigilance review.